GENERATION OF PERFECT MAP CODES FOR AN ACTIVE STEREO IMAGING SYSTEM

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Abstract

In the past, structured light (SL) methods typically relied on global reference frames to index points on the illumination pattern. This has limited their use. We are using perfect maps (PM) codes which can be labeled using local clues alone. A difficulty with PM has been their construction. Previously, we have relied on pseudorandom search techniques to identify suitable candidates. Here we give methods for direct construction of these two dimensional arrays. Specifically, a method is given for the construction of a *3*-ary (27,729; 3,3) Perfect Map. In order to achieve some error detection and correction, higher hamming distance codes are needed. A simple search algorithm is also given for the generation of such maps.

INTRODUCTION

A significant problems of stereo vision is identifying the corresponding features in each view [1]. To overcome this difficulty, artificial features can be introduced by illuminating the scene with structured light (SL)[2]. In this approach, one of two passive cameras of stereo vision is replaced by an active projector.

Most earlier SL applications used a line grid with the intersections points used for computing the 3D coordinates of the imaged object. To correctly index these patterns every line has to be identified. Occlusion of even a small part of the pattern can seriously compromise the indexing accuracy. Temporal codes have been proposed as solution to this problem. In their simplest form, a single point is rastered across the imaged object. A more complex form of these relies on binary spatiotemporal encoded dot patterns.

We have been using both spatial color PM codes and temporal monochrome PM patterns as an alternative which permit rapid image acquisition while allowing for identification of projection points from local cues. We have previously reported on successfully application of the technique to imaging inanimate objects [3], skin wound reconstruction [4] and to frozen brain surface reconstruction for sectional alignment [5].

The essence of the codes is illustrated in Fig. 1. Each element in the array unambiguously identified by its value and that of the nearest 8 neighbors. That is, every 3x3 window within the array is unique and is not repeated anywhere else in the map.

Construction of large PM has proven to be difficult using pseudorandom search techniques. Even more difficult has been identifying ones in which the minimum difference between the 3 x 3 windows, the Hamming distance (H), is large so that error detection and correction would be possible. Here we report on new superior techniques for generating these codes.

3	3	3	2	3	3	2	1	2
1	2	3	1	3	1	2	3	3
2	1	2	3	2	2	3	2	1
3	3	3	1	2	3	1	3	3
1	2	3	2	3	3	3	2	1
3	2	1	2	3	2	1	1	2
2	1	2	3	2	1	2	1	3

Figure 1. An example of portion of a perfect map. Each of the 3X3 windows is unique. Each numeral can correspond to a different color in the case of structured light imaging with the spatial color PM or, in the temporal implementation, to a time point, one of three different ones, the given point is illuminated.

DEFINITIONS

A m-ary de Bruijn Sequence of order n (m-ary dBS order n) is a circular sequence of length l ($l = m^n$) consisting from m letters where every window length n is unique and every possible m letter n-length code is present somewhere in the sequence [6].

E.g.: 2ary dBS order *3* = { 0 0 0 1 1 1 0 1}
unique windows of length 3 are:
{ 000, 001, 011, 111, 110, 101, 010, 100 }

A t-ary (M,N; m,n) Perfect Map (t-ary (M,N; m,n) PM) is a circular two-dimensional t letter M by N matrix in which every m by n submatrix is unique and all possible t letter m by n matrixes are present somewhere in the M by N matrix [7].

METHOD 1

Described here is an analytic method for constructing a PM with H=1. As an example, for the construction of a *3* ary (27,729;3,3) PM, we start with the one dimensional *3*-ary

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dBS order 3 as the first column: {0 0 0 1 1 1 2 2 2 0 0 2 2 1 1 0 1 2 0 2 1 2 1 0 2 0 1 }

If shifted versions of the first column are placed in the following columns and as long as each of the neighboring two shifts is unique in the shift sequence, we will have a two dimensional array with 3x3 window property. Since shifts can be of any value between 0 and 26, the question is how to generate a 27 letter shift sequence where every window of size two is unique? That is, we require *27*-ary dBS order *2*. The length of this code is 729 ($=27^2$). Generating such a code is not trivial. Below we give a sample method:

We can write the following row which has all the possibilities of windows of size two who have 0 (except 26 0).

r0 = {0 0 1 0 2 0 3 0 .....0 24 0 25 0 26}

similarly a row for *1* can be written:

r1 ={ 1 1 2 1 3 1 4 1 5 ..... 1 24 1 25 1 26}

We do not include (1 0) and (0 1) in *r1*, not to repeat the pairs already present in *r0*.. Similarly:

r2 = { 2 2 3 2 4 2 5 2 ..... 2 24 2 25 2 26}

.....

r24 = {24 24 25 24 26}

r25 = {25 25 26}

r26 = {26}

Please note that each of the rows above have a window property of two and together they use all the combinations of *27* letter length *2* codes except the following ones {(26 0), (26 1), (26 2).....(26 24), (26 25), (26 26)} If we put all rows side by side, all of these codes will be used as bridge codes and we generate a *27*-ary dBS order 2 or an 3-ary(1,729;1,3)PM={[r0],[r1],..[r26]}

We have now two sequences: a *3*-ary dBS order *3* and a *27*-ary dBS order *2*. We can generate a (27,729;3,3) PM as follows:

•first column C1 is *3*-ary dBS order 3

•other columns are shifted versions of column C1. The amount of shift of C(i) from previous column C(i-1) is i-1'th element of the *27*-ary dBS order *2*. (for i = 2,3,4....729)

METHOD 2

This method described here allows generation of higher hamming distance codes using an evolutionary algorithm. In this approach a random code of the desired size is generated and later changes are done according to the following procedure:

1. Select an element in a random location,
2. Make a random change,
3. Keep the change if the result is more desirable, if not, reverse the last change,
4. go to step 1

On the step 3, a cost function is used to evaluate the status of the array. For the hamming distance one case, the cost function is simply the number of windows which are repeated somewhere else in the array. In the higher hamming distance cases, the cost function is a weighted sum of the total number of undesired hamming distance encounters.

In order to generate this hamming distance histogram the array has to be converted to its corresponding index matrix by convolution with an indexing kernel. An indexing kernel is a matrix same size as the desired window and contains the increasing powers of the total available characteristic number. E.g.: for three letters it could look like A={1,3,9;27,81,243;729,2187,6561}.

After convolution every element of the resulting matrix should be compared with the remainder of the matrix by either sorting or using a lookup table.

DISCUSSION

Structured light imaging using PM has proven to be of great use. In both its spatial and temporal form, it allows faster imaging than binary temporal codes while the indexing problem is significantly simpler than that of other spatial codes. With the methods described here, it is now possible to readily generate these codes. The development of a technique for generating PM with high Hamming distance will allow incorporating error detection and correction into the imaging process.

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